

1 Voice processing — calls into records

Speech recognition	Open model + 20 h of 119 call training
Trained portion	Only 3.6% of the model (LoRA)
Extraction	Korean encoder + custom layers
Output	15 fields · always the same format

15 judges · E-Gen severe-disease groups — reperfusion, brain hemorrhage surgery, aortic emergency, severe burns + 11 more

Where it's used

- 119 EMS — the run report is written automatically from the call
- ER — pre-arrival calls go straight to the triage screen
- Regional dispatch — share once instead of re-explaining to 10 hospitals

2 Reliability model — the shelf life of information

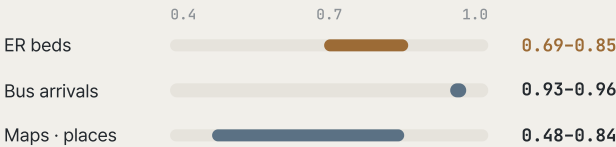
Lifespan learning never reads the content. It learns only **who** · **how often** · **how** a source has edited its data, and outputs the probability that it is stale right now.

Prior theory — “What changed often has probably changed again”
→ rank scores only, one rule for every source

GoldenLink — “Given this history, how likely is it stale now?”

On the same history, change-frequency methods show up to **168x** higher probability error (IBS)

Where it's used · validated in 4 domains



The 0.48 floor for maps · places is near random (0.5) — such data is screened out first by the **automatic diagnosis** (applicability check) before adoption.

AI search · RAG is validated by outcome, not discrimination — stale-evidence answers 33.3% → 23.3% (preliminary, single corpus)

Discrimination: 1.0 = perfect · 0.5 = random. Paper “Learning the Shelf Life of Information” submitted to an international journal.

We put an expiry date on data

The engine proven on ER beds, applied to any data that goes stale

Today, all data is refreshed on the same cycle and trusted equally until the next refresh. But every dataset goes stale at its own speed — even within one feed, item lifespans differ **8x**.

Information Reliability Engine

API · On-premise

IN	You send · data IDs and edit timestamps — no content or personal data
OUT	You get · probability it's correct now · expected lifespan · when to re-check
START	Before adoption · we start with an automatic diagnosis of whether it works on your data

What it does

- Tells you when to re-check** — 11% less time on stale data, same budget
- Filters before use** — low probability means verify first
- Audits reported data** — reports where agency-reported values go stale

Who it's for

Fire · dispatch · local gov't	Bed reliability + automatic call records (GoldenLink) — choose hospitals from trustworthy candidates
AI search · chatbot teams	Checks whether evidence is still valid before answering, re-verifying only the sources that need it — stale-evidence answers 33.3% → 23.3%
Data providers · platforms	Per-item re-check alerts + reports on where reported values go stale — 11% less time on stale data, same budget (median)

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INSTAGRAM

Delayed ER transfers,
a.k.a. ‘ER ping-pong’

Why do ambulances wander in search of a hospital?

2.4x

Two-year rise in transfers taking 30+ minutes from scene arrival to hospital departure



Is it really a lack of beds?

Open to find out. →

TEAM PLAYGROUND

AI FESTA 2026

01 · PROBLEM

Why won't ER ping-pong go away?

It's not a lack of beds —
it's a lack of **reliable data**

2×

ERs declaring they can't
accept patients
2023 → 2024

5.4%

Transfers taking over 30
min
up from 1.9% in 2023

Why it keeps happening

- Paramedics check hospitals **one phone call at a time** — even dispatch centers take **33 min** on average and contact **10 hospitals**
- The top reason for refusal isn't beds but **staff shortages (48.1%)** — a defensive call made without information

With reliable information, the first thing to drop is 'refuse because we don't know.'

People tried to fix it — but there was no reliable data

The government is speeding up allocation, but when we cross-checked the hospital data behind those decisions:

91.2%

of hospitals that said 'cannot accept'
actually had a specialist in that department

8.6 yrs

Max age of stale bed values shown as real-time (25 hospitals)

0 → 4

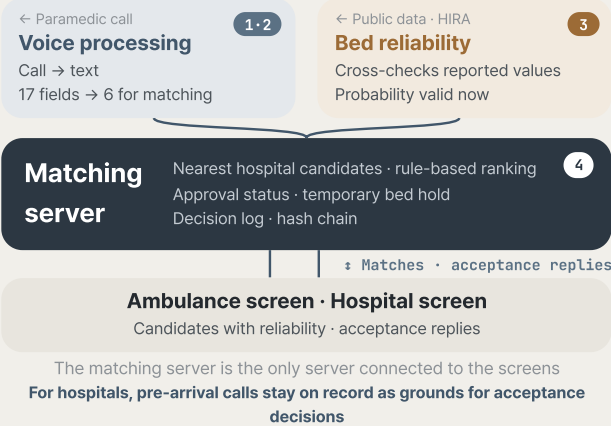
Burn-center capacity — 0 found from a single source, 4 when cross-checked

Bed data is entered manually by staff. Having a specialist does not guarantee acceptance. Sources: National Medical Center, National Fire Agency, National Assembly; team measurements (E-Gen, HIRA).

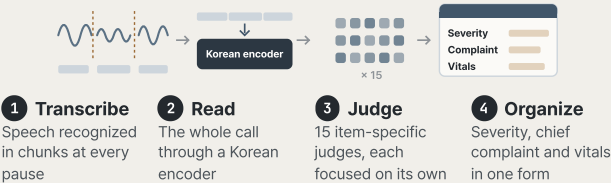
02 · SO HERE'S HOW WE SOLVED IT

One call, evidence-backed
candidates

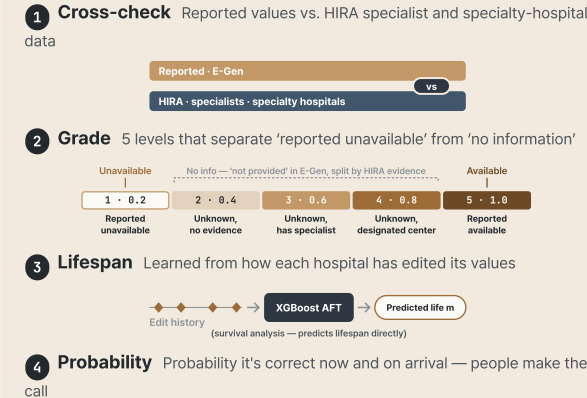
Calls are organized automatically; hospital data is verified.



1 Voice → patient info · how it works



2 Bed reliability · how it works

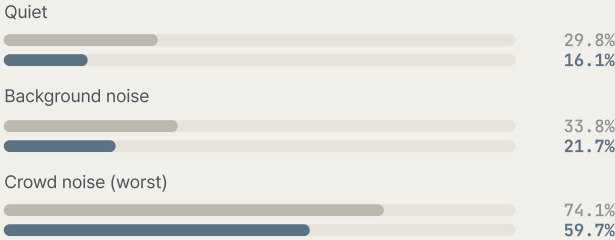


03 · PERFORMANCE — MEASURED OURSELVES

Results in numbers

1 Speech recognition · character error rate

500 sentences from real 119 calls



2 Patient info extraction

	Severity	Complaint	Per case
GoldenLink 0.35B	75%	97%	0.013 s
Generative 1.7B	78%	94%	6.9 s
External LLM API	61%	85%	0.4 s+

0.01 s with no data sent outside — accuracy on par with large models

3 Bed reliability validation

Live link to ERs nationwide ^⓪	533
Merged with HIRA · median location error 11 m	97.2%
Time spent on stale data · same budget	−11%
Stale evidence in AI answers ^⓪	33.3→23.3%

⓪ 3 of 7 noise conditions shown ⓪ 100 synthetic calls, same scoring ⓪ median of retrospective ER-bed simulation. ⓪ 533 = all ERs with live E-Gen bed feeds (as of ⓪) ⓪ AI answers n=⓪⓪, preliminary single-corpus result. All figures before human review.